

FRIDA

Frida LAB 6

Objective

In Challenge 0x6, the application validates the return value of a private method. Unlike previous challenges, the method is not publicly accessible, but Frida can still hook and replace its implementation at runtime.

The goal is to hook the private method and force it to return the expected value so the application reveals the flag.

Install the APK

Install the application on your rooted Android device or emulator.

```
adb install FridaLab.apk
```

Launch the Application

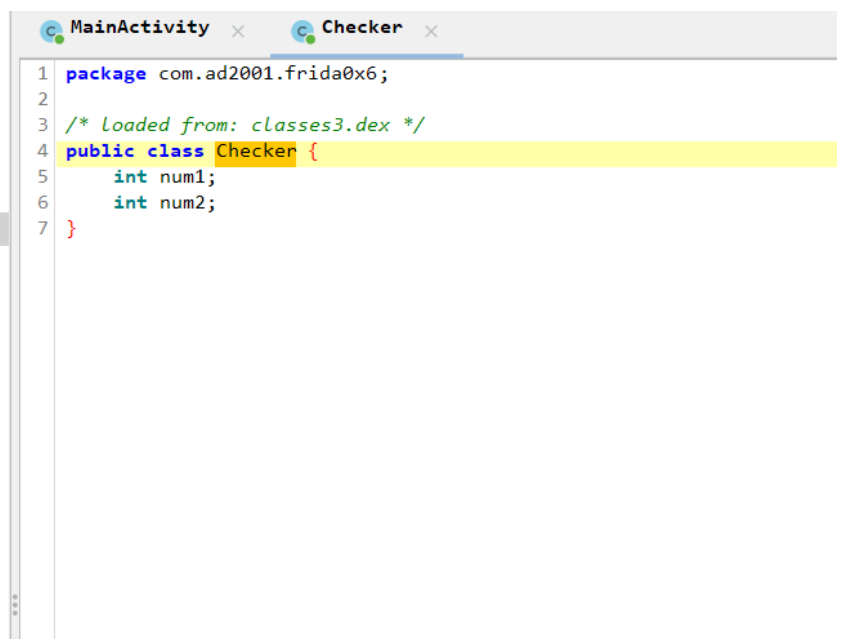
Decompile the APK

Open the APK in JADX.

```
jadx-gui FridaLab.apk
```

Browse the source code.

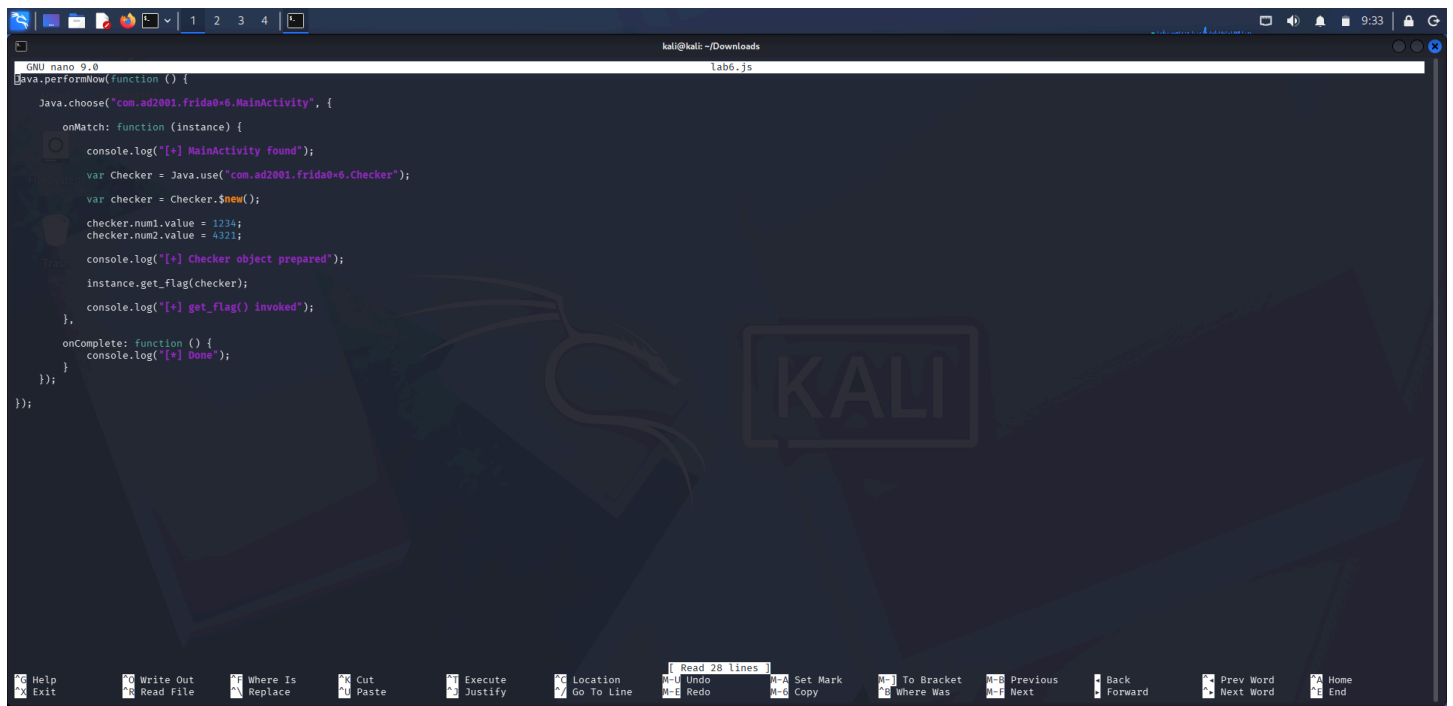
```
debug.apk
source code
android.support.v4
androidx
com
  ad2001.frida0x6
    Checker
      num1 int
      num2 int
    MainActivity
    R
  google
  kotlin
  kotlinx.coroutines
  org
  sources
  K signature
  mmary
```



```
1 package com.ad2001.frida0x6;
2
3 /* loaded from: classes3.dex */
4 public class Checker {
5     int num1;
6     int num2;
7 }
```

Write the Frida Script

Create a file.



```
GNU nano 9.0 lab6.js
Java.performNow(function () {
  Java.choose("com.ad2001.frida0x6.MainActivity", {
    onMatch: function (instance) {
      console.log("[+] MainActivity found");
      var Checker = Java.use("com.ad2001.frida0x6.Checker");
      var checker = Checker.$new();
      checker.num1.value = 1234;
      checker.num2.value = 4321;
      console.log("[+] Checker object prepared");
      instance.get_flag(checker);
      console.log("[+] get_flag() invoked");
    },
    onComplete: function () {
      console.log("[+] Done");
    }
  });
});
```

D:\Downlaods> adb shell

generic_x86:/ # cd /data/local

local.prop local/

generic_x86:/ # cd /data/local/tmp/

generic_x86:/data/local/tmp # ls

generic_x86:./frida

Run Frida

Start the application with Frida.

frida -U -f uk.rossmarks.fridalab -l challenge6.js

```
kali@kali:~/Downloads$ cat lab6.js
87-82 11:48:21.979 238 238 0 |init| : fd_exec_cmd am broadcast -a com.lidmng.APP_START -p com.android.1d.spsptore -f 32 --es package com.sadms.frida04 --es pos pc
87-82 11:48:21.131 238 238 0 |init| : fd_exec_cmd am broadcast -a com.lidmng.APP_START -p com.lidmng.launcher3 -f 33 --es package com.sadms.frida04 --es pos pc
87-82 11:48:21.180 825 3780 0 |AppStart|:action 1 com.lidmng.APP_START package name = com.sadms.frida04 pos = pc

kali@kali:~/Downloads$ frida-ps -ui
PID Name Identifier
4388 Frida 0x5 com.sadms.frida04
5236 Frida 0x5 com.sadms.frida04
2572 Google Play Store com.android.vending
5238 LIDStore com.android.1d.spsptore
825 Launcher3 com.lidmng.launcher3
777 Settings com.android.settings
- Chrome com.android.chrome
- Dms jakhu.spsm.dms
- Facebook com.facebook.katana
- Files com.android.documentsui
- Frida 0x2 com.sadms.frida04
- Frida 0x3 com.sadms.frida04
- Frida 0x4 com.sadms.frida04
- Gallery com.android.gallery3d

kali@kali:~/Downloads$ frida -u -p 5236 -l lab6.js
$ frida -u -p 5236 -l lab6.js

Frida 17.15.3 - A world-class dynamic instrumentation toolkit

Commands:
  help          -> Displays the help system
  object        -> Display information about "object"
  exit/quit    -> Exit

- - -
- Prefer a GUI? Luma is the official Frida app, with a live REPL.
- persistent sessions & collaboration. https://luma.frida.re/
- - -
Connected to HD1918 (id=192.168.1.48:5555)

Attaching...
[*] MainActivity found
[*] Launcher object prepared
[*] get_flag() invoked
[*] Done
[HD1918::PID::5236] ->
```

Final output look likes :

```
frida_Lab6_Report.pdf
LIDPlayer | Frida 0x3 | Frida 0x4 | Frida 0x5 | Frida 0x6
11:49
@kali:~/Downloads$ cat lab6.js
87-82 11:48:21.979 238 238 0 |init| : fd_exec_cmd am broadcast -a com.lidmng.APP_START -p com.android.1d.spsptore -f 32 --es package com.sadms.frida04 --es pos pc
87-82 11:48:21.131 238 238 0 |init| : fd_exec_cmd am broadcast -a com.lidmng.APP_START -p com.lidmng.launcher3 -f 33 --es package com.sadms.frida04 --es pos pc
87-82 11:48:21.180 825 3780 0 |AppStart|:action 1 com.lidmng.APP_START package name = com.sadms.frida04 pos = pc

kali@kali:~/Downloads$ frida-ps -ui
PID Name Identifier
4388 Frida 0x5 com.sadms.frida04
5236 Frida 0x5 com.sadms.frida04
2572 Google Play Store com.android.vending
5238 LIDStore com.android.1d.spsptore
825 Launcher3 com.lidmng.launcher3
777 Settings com.android.settings
- Chrome com.android.chrome
- Dms jakhu.spsm.dms
- Facebook com.facebook.katana
- Files com.android.documentsui
- Frida 0x2 com.sadms.frida04
- Frida 0x3 com.sadms.frida04
- Frida 0x4 com.sadms.frida04
- Gallery com.android.gallery3d

kali@kali:~/Downloads$ frida -u -p 5236 -l lab6.js
$ frida -u -p 5236 -l lab6.js

Frida 17.15.3 - A world-class dynamic instrumentation toolkit

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Attaching...
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```